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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: _____ Date: 30/8/2025

Abstract

This research introduces a prenatal health monitoring system that supports pregnant mothers with continuous and accessible care. The system collects important health parameters such as blood pressure, heart rate, glucose level, body mass index (BMI), fetal heart rate, and uterine contractions. These readings are processed through a machine learning model to detect possible risks earlier and more accurately than traditional single-factor methods.

The system has four main features. The Vital Signs Monitoring feature allows mothers to record and track their key health readings. The Early Warning System identifies complications such as preeclampsia, gestational diabetes, abnormal weight gain, and preterm labor. It provides real-time alerts with simple explanations, delivered through a mobile application or SMS to both mothers and doctors. The Personalized Health Insights feature offers weekly trimester-based tips on rest, nutrition, activity, and stress management to guide mothers in daily routines. Finally, the Visual Health Score Dashboard shows weekly health trends using color codes, gives clear explanations of changes, and creates exportable summaries that can be shared with doctors for better medical decisions.

The novelty of this system is the integration of multiple health indicators into a machine learning model that adapts to each mother's condition. Unlike traditional approaches, this method gives more personalized and reliable predictions. The user-friendly dashboard and explainable outputs help both mothers and doctors to make better decisions. By combining early detection, real-time alerts, and personalized insights, this system aims to improve maternal and fetal health outcomes in a simple and effective way.

Keywords: prenatal care, health monitoring, machine learning, pregnancy, maternal health

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1. INTRODUCTION

1.1 Background & Literature Survey

Background

Pregnancy is one of the most important phases in a woman's life, where both the mother and the baby's health need careful attention. Maintaining good health during this period is crucial, as various complications can arise that may pose serious risks if not detected early. Common problems during pregnancy include high blood pressure, gestational diabetes, unusual weight changes, and preterm labor. If these issues are not monitored properly, they can lead to severe health consequences for both the mother and the baby.

Traditional prenatal care often relies on occasional visits to hospitals or clinics, where only a limited set of health parameters are checked. This approach may not provide a comprehensive understanding of the mother's overall health. As a result, certain complications may go unnoticed until they become critical, increasing the risk of adverse outcomes.

With the advancement of technology, it is now possible to improve prenatal care by continuously monitoring maternal health using mobile applications and smart devices. These tools can record vital health parameters such as blood pressure, heart rate, blood glucose levels, body mass index (BMI), fetal heart rate, and uterine contractions. By collecting this information, both mothers and healthcare providers can track health trends, receive timely alerts about potential risks, and access personalized recommendations to maintain well-being.

A system that integrates all these features into a single platform can significantly enhance prenatal care. Such a system can display the collected data in an easy-to-understand dashboard, provide personalized health insights, and issue early warning alerts when complications are detected. This helps mothers take better care of themselves at home and enables doctors to make informed clinical decisions. Ultimately, a data-driven, user-friendly prenatal health monitoring system has the potential to make pregnancy safer and improve maternal and fetal outcomes.

Literature Survey

Many researchers have tried to improve monitoring of a mother's health during pregnancy. The World Health Organization and the American College of Obstetricians and Gynecologists emphasize that blood pressure and blood sugar are critical indicators for identifying complications such as preeclampsia and gestational diabetes [1][2]. While these methods are effective in detecting certain risks, they only look at one or two factors. As Smith discussed, this makes it difficult to understand the complete health condition of both the mother and the baby [7]. For example, a mother's blood pressure may appear normal, but if her glucose levels or body weight are abnormal, serious risks may still be missed.

To address these limitations, researchers have explored advanced technologies. Brown explained that artificial intelligence can process large amounts of maternal health data and detect patterns more accurately than traditional methods [3]. Kumar highlighted that machine learning models which analyze multiple health indicators together, such as

glucose, blood pressure, and BMI, are able to detect complications at an earlier stage [4]. For instance, gestational diabetes may not be noticed if only blood pressure is measured, but when glucose levels and BMI are analyzed together, the risk becomes much clearer.

Some researchers have also focused on real-time monitoring and alert systems. Johnson introduced an AI-based solution that provides mothers and doctors with personalized advice and immediate alerts when abnormal values are detected [6]. Similarly, Gupta's work showed that intelligent prenatal care systems can prevent complications by warning healthcare providers early [8]. These systems have shown promise in improving safety and decision-making. However, most of these applications focus mainly on maternal health and do not include direct fetal health indicators such as fetal heart rate or uterine contractions, which are equally important.

In addition, several studies have pointed out usability issues. Lee argued that many monitoring tools only track a few parameters and do not provide a clear and simple dashboard for mothers and doctors [5]. Wang also noted that in many systems, the data is fragmented and difficult to interpret, which leads to delayed actions [9]. This gap is important because if mothers cannot understand their health trends easily, the benefits of monitoring are reduced.

The proposed system in this research is designed to overcome these gaps. It will combine both maternal and fetal health data in one platform. The chosen health indicators include blood pressure, heart rate, glucose, BMI, fetal heart rate, and uterine contractions. The system will generate real-time alerts for any abnormal conditions, provide weekly personalized health tips for mothers, and present the results in a clear visual dashboard that highlights health trends over time. Unlike many existing systems, this approach is more complete and easier to use. It supports mothers in taking proactive steps for their safety while also giving doctors a better decision-making tool.

1.2 Research Gap

Previous studies on prenatal health monitoring have often focused on only one or two health measurements, such as blood pressure or blood sugar [1][2]. While these single-parameter methods can help detect some problems, they may miss other important signs of complications, which can lead to delayed detection and increase risks for both the mother and the baby. For example, a mother may have normal blood pressure but abnormal glucose levels or fetal heart rate, which could indicate a hidden risk.

Some research has attempted to combine multiple health indicators using advanced methods [3][4]. These studies show that monitoring several parameters together can improve the accuracy and timeliness of detecting potential complications. However, many existing systems are not very user-friendly for mothers or doctors. They often lack clear dashboards, simple visualizations, or explanations of the results, making it difficult for users to understand the overall health trends [5][9].

Other systems provide alerts or advice based on recorded measurements [6][8]. While these features are helpful, they may not cover all critical health factors, and the guidance is often generic rather than personalized according to each mother's trimester. This

reduces the effectiveness of the system in real-life situations, where timely and tailored advice is crucial [10][11].

From this review, it is clear that there is a strong need for a comprehensive prenatal health monitoring system that:

- Monitors multiple maternal and fetal health parameters (blood pressure, heart rate, glucose, BMI, fetal heart rate, and uterine contractions).
- Provides real-time alerts whenever any parameter goes beyond safe limits.
- Gives personalized guidance for each mother depending on her trimester.
- Displays information through a clear and easy-to-understand dashboard so both mothers and doctors can quickly identify risks.

The novelty of the proposed system lies in combining these elements in one integrated platform. Unlike many existing systems, this system:

1. Application novelty – Provides a complete solution for home monitoring and clinical support, combining maternal and fetal parameters in a user-friendly, practical way.
2. Technical novelty – Implements rule-based monitoring, trend visualization, and real-time alerts for multiple parameters, allowing early detection of risks and easy interpretation of health data.

This approach ensures mothers can manage their health at home, doctors receive a full overview of maternal and fetal well-being, and timely actions can be taken to improve outcomes [5][6][7][9][10].

1.3 Research Problem

Pregnancy is a critical period during which both maternal and fetal health must be closely monitored. Many women experience complications such as high blood pressure, gestational diabetes, abnormal weight changes, and preterm labor. If these complications are not detected early, they can lead to serious health risks for both the mother and the baby [1][2]. Traditional prenatal care usually relies on periodic hospital visits and focuses on a few health measurements, which may not provide a complete picture of the mother's overall condition [3][7].

Some existing health monitoring systems have attempted to address this issue. For example, single-parameter monitoring systems track only one factor, such as blood pressure or blood sugar, which may miss other signs of complications [1][2]. While some advanced systems use AI or machine learning to combine multiple health indicators for early detection [4][5][6][8], many of these systems are not user-friendly. They often lack clear dashboards, simple visualizations, or personalized guidance that is specific to each mother's condition and trimester [5][9][13].

These limitations make it difficult for mothers to understand their own health status and for doctors to make timely and informed decisions. There is a clear need for a **comprehensive prenatal health monitoring system** that:

1. Tracks multiple maternal and fetal health parameters (blood pressure, heart rate, glucose, BMI, fetal heart rate, and uterine contractions),
2. Provides real-time alerts when abnormal conditions are detected,
3. Offers weekly personalized guidance tailored to the mother's trimester, and
4. Displays information in a simple, clear, and visual dashboard that is easy to understand.

Such a system would help mothers manage their health at home, support doctors with more complete information, reduce risks, and improve overall maternal and fetal outcomes [5][6][7][10][12][14].

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to develop a comprehensive and easy-to-use prenatal health monitoring system that helps pregnant mothers and doctors keep track of important health measurements. The system aims to detect any early signs of complications and provide timely alerts, so that both mothers and healthcare providers can take the necessary actions quickly. In addition, it seeks to offer personalized guidance and health advice based on the mother's condition and the stage of pregnancy. By combining continuous monitoring, early warnings, and simple advice in a single platform, the system intends to improve the overall safety and well-being of both mothers and their babies during pregnancy.

2.2 Specific Objectives

Monitor Multiple Health Parameters

To record and track important health measurements of pregnant mothers, including blood pressure, heart rate, blood sugar, BMI, fetal heart rate, and uterine contractions. This will give a complete view of the mother's condition at any time.

Early Detection of Complications

To identify early signs of pregnancy complications, such as high blood pressure, gestational diabetes, abnormal weight changes, and preterm labor. The system will provide real-time alerts with simple explanations to help mothers and doctors take timely action.

Provide Personalized Health Advice

To give weekly guidance based on the mother's health condition and trimester. This includes tips on rest, nutrition, physical activity, and stress management, helping mothers follow safe practices throughout pregnancy.

Visualize Health Trends

To display health data in a clear, easy-to-understand dashboard. The dashboard will show trends, highlight any risks, and allow doctors to review summaries for better decision-making.

Support Maternal and Fetal Health

To improve overall health outcomes by combining monitoring, early alerts, personalized advice, and clear visual information in a single, user-friendly system.

3. METHODOLOGY

The proposed prenatal health monitoring system will be developed using a step-by-step approach to ensure it is effective, accurate, and easy to use.

1. Requirement Analysis

First, the system requirements will be identified by studying what is needed for effective prenatal monitoring. This includes:

- **Health Parameters:** Decide which measurements to track, such as blood pressure, heart rate, glucose level, BMI, fetal heart rate, and uterine contractions.
- **Alerts:** Determine what types of alerts are needed, such as high blood pressure, abnormal glucose, or unusual fetal heart rate.
- **Personalized Advice:** Identify the kind of weekly guidance mothers need, including tips on rest, nutrition, exercise, and stress management, based on their condition and trimester.
- **Dashboard Features:** Decide how health trends, alerts, and summaries should be displayed clearly for both mothers and doctors.

2. Data Collection

Health data such as blood pressure, heart rate, glucose level, BMI, fetal heart rate, and uterine contractions will be collected. This can be done through manual entry by the mother or by integrating wearable devices and smart health monitors.

3. System Design

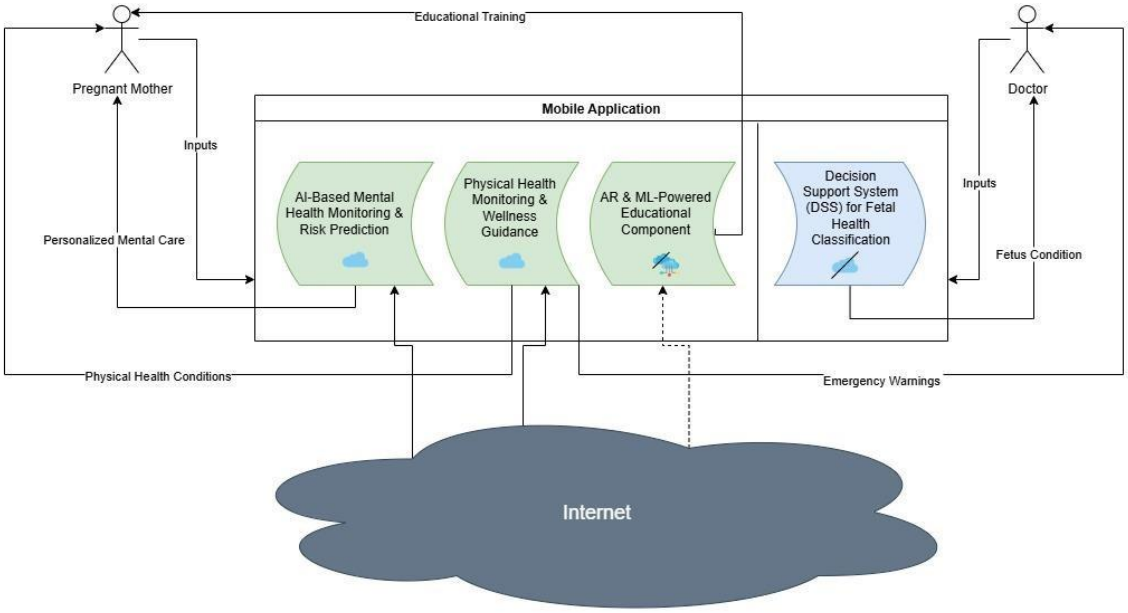

Figure 1- Overall System Diagram

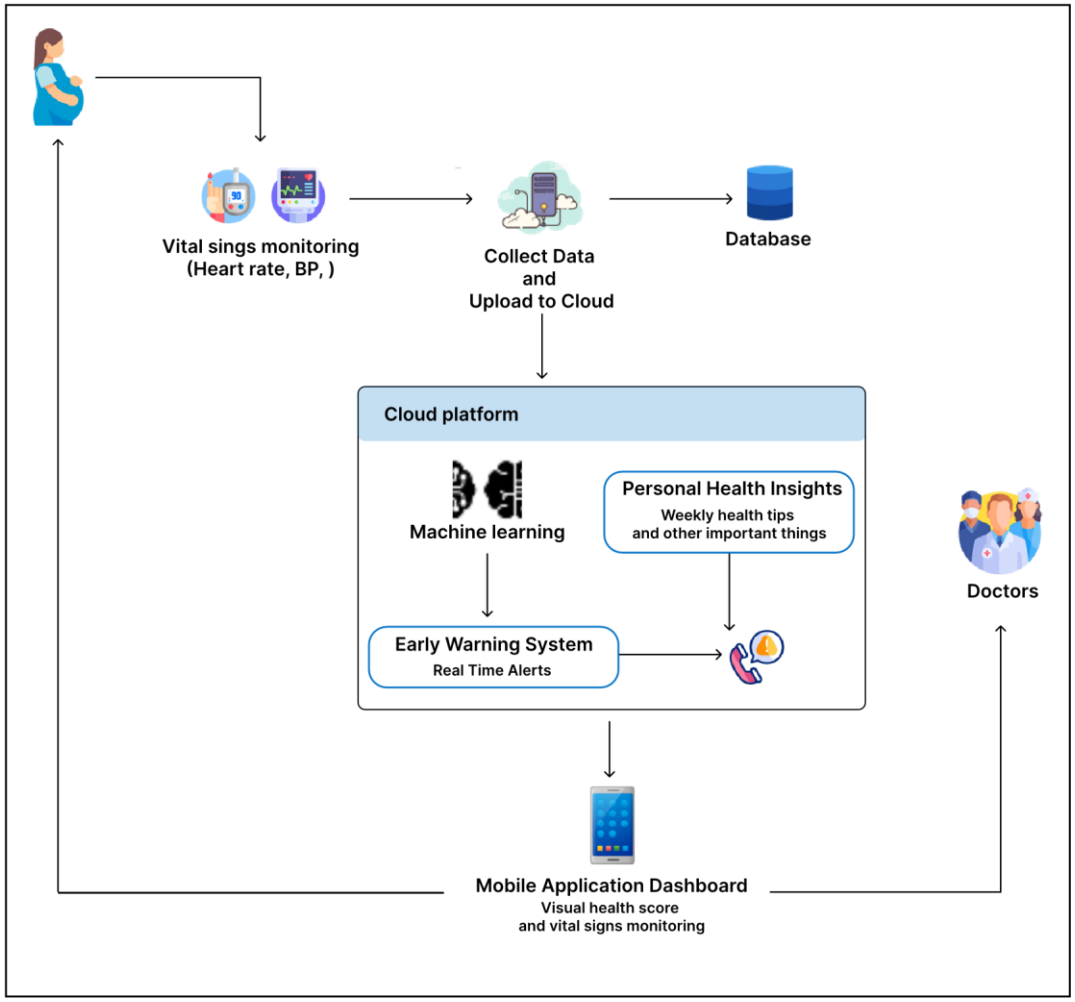

Figure 2 - Component System Diagram

The system will be designed with four main modules:

- Vital Signs Monitoring: To record and track all health measurements.
- Early Warning System: To detect signs of complications and send alerts.
- Personalized Health Insights**:** To provide weekly health tips based on the mother's condition and trimester.
- Visual Health Score Dashboard: To show trends in a clear and simple way, highlighting risks and summaries for doctors.

4. Machine Learning Model
A machine learning model will be implemented to analyze the collected health data. It will learn from patterns in the data to detect possible complications early and provide accurate predictions.

5. System Implementation
The system will be developed as a mobile and/or web application using appropriate programming tools. It will include data entry interfaces, alert notifications, advice modules, and the dashboard for visualization.

6. Testing and Validation
The system will be tested using sample data to check accuracy, reliability, and usability. Feedback from doctors and mothers will be used to improve the system before final deployment.

7. Evaluation
Finally, the system's performance will be evaluated by measuring how accurately it detects health risks and how effectively it helps mothers follow personalized advice.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

Vital Signs Recording

- Pregnant mothers can input and update key health parameters: blood pressure, heart rate, glucose, BMI, fetal heart rate, and uterine contractions.
- The system stores historical health data for trend analysis.

Early Warning System

- Detect early signs of complications such as preeclampsia, gestational diabetes, abnormal weight gain, and preterm labor.
- Send real-time alerts to mothers and doctors via app or SMS.
- Provide explanations with each alert to guide actions.

Personalized Health Insights

- Generate weekly health tips based on the mother's condition.
- Adjust recommendations according to the trimester, covering rest, nutrition, activity, and stress management.

Visual Health Score Dashboard

- Display weekly health trends with color-coded indicators.
- Show key factor explanations for each parameter.
- Allow exporting health summaries for doctors in PDF format.

Machine Learning Integration

- Combine multiple health indicators into a predictive model for early detection of complications.
- Continuously improve prediction accuracy using adaptive learning.
- Provide explainable AI outputs to support clinical decisions.

User Management

- Support secure login and profile management for mothers and doctors.
- Enable doctors to access patient data with consent.

4.1 Non-Functional Requirements

Performance

- Alerts should be generated and sent quickly after detecting a risk.
- Dashboard updates and data visualizations should load without noticeable delay.

Usability

- The interface should be user-friendly and easy to navigate.
- The system should be accessible via mobile and web, even in low-resource settings.

Reliability & Availability

- The system should maintain high availability.
- Data should be backed up regularly.

Security & Privacy

- All data must be encrypted during storage and transmission.
- The system must comply with healthcare data privacy regulations (e.g., HIPAA/GDPR).
- Multi-factor authentication should be used for mothers and doctors.

Scalability

- The system should support an increasing number of users without performance issues.
- The machine learning model should handle large datasets as usage grows.

Maintainability & Extensibility

- The system should allow easy updates to ML models and health parameter thresholds.
- The design should be modular to allow adding future features, such as nutrition tracking or remote consultations.

5. BUDGET AND BUDGET JUSTIFICATION

Table 1 - Budget Justification

Requirement	Cost Approx. (Rs.)
Travelling Cost	9000.00
Internet Usage Cost	11000.00
Contingency Cost	4000.00
Total Cost	**24000.00**

6. FAILURE MODE ANALYSIS

Table 2 - Failure Analysis

Failure Mode	Effect	Contingency Level
User enters incorrect health data	Alerts or insights may be inaccurate, leading to wrong decisions	High – System should validate input and allow corrections
Early Warning System fails to detect risk	Mother may not get timely alerts, increasing risk of complications	High – Regular testing and backup algorithms, manual review possible
Machine Learning model gives wrong prediction	Incorrect health risk assessment or recommendations	Medium – Periodic model retraining, threshold adjustment, manual override
Dashboard fails to load or display trends	Users cannot see health trends, affecting understanding and decision-making	Medium – Local caching of data, alternate visualization options
Personalized Health Insights are not generated	Mother does not receive weekly guidance, affecting preventive actions	Medium – Default generic tips can be displayed if algorithm fails

7. GANTT CHART

Table 3 - Gantt Chart

Tasks	Sep	Oct	Nov	Dec	Jan	Mar	Apr	May
Collect and preprocess prenatal health datasets								
Feasibility study & finalize datasets to be used								
Train ML models for anomaly detection								
Select and fine-tune pre-trained models								
Design risk scoring system								
Develop alert module for complications								
Frontend development start								
Weekly recommendation engine implementation								
Visual dashboard design								
Integrate ML models + fusion algorithm with backend								
Implement real-time alerts into frontend								
Continue frontend dashboard development								
Security & privacy implementation								
Functional testing								

Pilot study with small group								
Collect feedback on usability & accuracy								
Final integration of all modules (backend + frontend + ML models)								
System-wide testing								
Prepare documentation (progress + final report draft)								
Final Report								
Final Viva Presentation								

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